

No. of Functional Features Included in the Solution

Date	15 July 2024
Team ID	SWTID-2026-3798
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module/Component		Marks Contribution
1	Data Preprocessing	Clean & scale data	Preprocessing	Done	High
2	EDA	Analyze dataset	EDA Notebook	Done	Medium
3	Model Training	Train XGBoost model	ML Model	Done	High
4	Model Prediction	Predict flood risk	Prediction	Done	High
5	Flask Backend	Handle requests	Flask App	Done	High
6	User Interface	HTML/CSS frontend	Templates	Done	Medium
7	Model Saving	Save model & scaler	Joblib	Done	Medium
8	Result Display	Show prediction result	Flask UI	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home, Prediction Pages
2	Backend / Logic	3	Flask Routes, Prediction
3	Database / Storage	1	Saved Model(.save)
4	API / Integration	1	Flask Integration
5	Security / Authentication	0	Not Implemented